ISO 9001:2015 14001:2015 45001:2018 PED CERTIFIED

NPFI

MILL TEST CERTIFICATE

EN 10204:2004 3.1 / ISO 10474:1991 3.1

Manufacturer of Pipes, Fittings & Flanges in Carbon Steel, Alloy Steel, Stainless Steel, Duplex Steel & Nickel Alloys

Customer	N-PIPE SOLUTIONS INC.	**Certificate No.**	NPFI/QC/2025/25059	**Date**	15-10-2025
P.O. No. / P. R. No.	P05006449	**Date**	06-10-2025		
Project Name	-	**Other Reference**	XXXX XXXX XXX XX		
Material Specification	ASTM A234-2018a GR. WPB	**Starting Material**	SEAMLESS PIPE		
Additional Requirement	NACE MR0175/MR0103	**Heat Treatment**	NORMALIZED - 920° C		

ITEM	DESCRIPTION	CONSTRUCTION	DIMENSION	SIZE				QTY.	HEAT NO.	RAW MATERIAL
NO.		TYPE	STANDARD	OD 1	WT 1	OD 2	WT 2	PCS.		
50	ELBOW 90D LR, BW R=1.5D	SEAMLESS	ASME B16.9-18	2" NB	SCH 80	-	-	1	N480	F1287

CHEMICAL COMPOSITION

ITEM	HEAT NO.	%	C	Mn	P	S	Si	Cr	Mo	Ni	Cu	V				F1	CEQ
		min.	--	**0.29**	--	--	**0.10**	--	--	--	--	--				--	--
		max.	**0.30**	**1.06**	**0.050**	**0.058**	--	**0.40**	**0.15**	**0.40**	**0.40**	**0.08**				**1.00**	**0.50**
50	N480	H	0.16	0.61	0.019	0.014	0.18	0.120	0.054	0.069	0.061	0.002				0.30	0.31
		P	0.15	0.60	0.018	0.013	0.17	0.130	0.055	0.068	0.060	0.002				0.31	0.30

F1 = Cu+Ni+Cr+Mo ; CEQ = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15

MECHANICAL PROPERTIES on Raw Material √ on Finished Material

TENSILE PROPERTIES									HARDNESS				IMPACT PROPERTIES					Client
TEST METHOD	ASTM A370						ROOM TEMP.						TEMPERATURE					Item Code
ITEM	HEAT NO.		O	S	YS	TS	EL	RA	HB				SPECIMEN	TEST RESULTS (Joules)				
NO.					MPa	MPa	%	%		(1)	(2)	(3)	Size (mm)	(1)	(2)	(3)	AVG.	
		min.			**240**	**415**	**30**	--		--								
		max.			--	--	--	--		**197**								
50	N480	P	L	R	306.69	492.28	41.35		B	143	144	144						30029764

We hereby certify that the material herein manufactured, sampled, tested & inspected in accordance with the above specification and requirements of Purchase order.

Notes:

Visual & Dimension : Satisfactory

All Fittings Bevel End DP tested : Satisfactory

as per ASTM E165

Legends :

O	Orientation	**S**	Strip		
H	Heat	**R**	Round	**RA**	Reduction Area
P	Product	**W**	Weld	**N**	Normalized
L	Londitudinal	**YS**	Yield strength	**N&T**	Normalize & Tempered
T	Transverse	**TS**	Tensile strength	**Q**	Quenched
B	Base	**EL**	Elongation	**ANL**	Annealed

Remarks:

All material supplied is certified to be free from mercury contamination & no mercury bearing equipment was used during manufacturing.

Killed steel & fine grain.

No weld repair was performed.

☐ Reviewed by :

☐ Witnessed by : _____

Auth. Signatory

Incharge (QA / QC Dept.)

Redg. Office: 42/46, Guru Rajendra House, 15/1St Floor, 5th Kumbharwada, Mumbai - 400 004, Maharashtra, India. Tel.: +91-22-23873829, Fax: +91-22-23824659

Works: Unit No. 16, 17 & 78, Choudhary Compound, Wakan Pada, Village - Dahanu Road, Nalasopara (East), Thane - 401209, Maharashtra, India